

Yashas Bhat

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Education

University of Maryland

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.57/4.0

College Park, MD

Aug. 2023 – May 2027

Technical Skills

Languages: Python, Rust, TypeScript, JavaScript, Java, SQL (Postgres), C/C++, OCaml

Frameworks & Tools: FastAPI, Flask, Axum, Next.js, React, Node.js, Tokio, FastMCP, Typer, LlamaIndex, Chroma, Git, Docker, Linux/UNIX, AWS

Experience

Amazon Web Services (AWS)

June 2026 – September 2026

Software Development Engineer Intern — Nitro SSD

Cupertino, CA

- Working on the Nitro SSD team to reduce turnaround time for shipping SSD software and firmware releases across the Nitro hardware stack.
- Contributing to low-level systems software in C/C++ on Linux for storage infrastructure underpinning AWS compute.

Children's National Hospital

August 2024 – May 2025

Student Developer

College Park, MD

- Architected **TrachSense**, a real-time patient monitoring platform for airway alerts for tracheostomy patients, and led development of the backend system behind **TrachHub**.
- Designed the real-time data pipeline and WebSocket streaming layer from sensor ingestion through mobile delivery, reducing monitoring latency to under 1 second.
- Developed a breathing-pattern analysis engine using hysteresis, RMS, standard deviation, and flatline ratio thresholds to classify breathing states and reduce false alerts by 10%.

Bharat Petroleum Corporation Limited

Dec. 2024 – Jan 2025

Software Engineering Intern

Mumbai, India

- Developed an anomaly detection framework combining DBSCAN clustering and Isolation Forest, identifying 950+ potentially fraudulent distributors from 320K+ digital customer verification transactions.
- Designed a feature extraction pipeline with 6 behavioral features, applying PCA visualization to surface irregular distributor behavior and weekday fraud trends for auditors.

Projects & Open Source

Canvas MCP | Python, FastMCP, Canvas API, Typer

July 2025

- Created an MCP server and CLI for Canvas LMS that lets AI clients query live course data across assignments, discussions, files, grades, modules, and calendar events.
- Implemented 25+ Canvas tools, URL resolution, and course-context snapshots with Chrome-cookie, session-cookie, and API-token auth across MCP transports.

Codexport | Rust, Axum, Rusqlite, Comrak, Tokio

March 2026

- Developed a Rust CLI and local web app for browsing Codex chat history, rendering sessions as transcripts or clean Markdown, and exporting conversations for reuse.
- Added print-friendly PDF views and expiring share links through a Tailscale Funnel share server.

Testudot | Python, FastAPI, Typer, BeautifulSoup, Docker

Dec. 2025

- Launched a Dockerized FastAPI service for monitoring UMD Testudo courses, detecting schedule changes and sending HTML email alerts with local and Redis-backed persistence.

Know | Python, LlamaIndex, Chroma

Jan. 2026

- Created a local search CLI that watches directories, chunks 20+ file types, and stores a persistent Chroma index with incremental indexing and BM25/hybrid retrieval (67% faster re-indexes).